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Microbes of the human eye: Microbiome, antimicrobial resistance and biofilm formation

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ABSTRACT

Background: The review focuses on the bacteria associated with the human eye using the dual approach of detecting cultivable bacteria and the total microbiome using next generation sequencing. The purpose of this review was to highlight the connection between antimicrobial resistance and biofilm formation in ocular bacteria.

Methods: Pubmed was used as the source to catalogue culturable bacteria and ocular microbiomes associated with the normal eyes and those with ocular diseases, to ascertain the emergence of anti-microbial resistance with special reference to biofilm formation.

Results: This review highlights the genetic strategies used by microorganisms to evade the lethal effects of anti-microbial agents by tracing the connections between candidate genes and biofilm formation.

Conclusion: The eye has its own microbiome which needs to be extensively studied under different physiological conditions; data on eye microbiomes of people from different ethnicities, geographical regions etc. are also needed to understand how these microbiomes affect ocular health.

1. Introduction

In 1676, Antonie Van Leeuwenhoek, an amateur microscopist from Holland, first discovered bacteria. Since then, bacteria from every possible niche in the environment (biotic and abiotic) including on the surfaces and those within the human body have been discovered. The major challenge in this endeavour of discovery were in isolating, cultivating and taxonomically identifying bacteria as well as in establishing their attributes (beneficial or harmful). These challenges were intrinsic to many bacteria themselves, as only about 1% of the total community was amenable to cultivation. With the advent of Next generation sequencing (NGS), culture-independent methods to detect microbes have become extremely popular and have led to the identification of several bacteria that were hitherto not culturable in the laboratory. This method does not rely on culturing of microbes for identification; instead NGS depends on a researcher's ability to extract bacterial DNA from a biological source. From this DNA, amplified genes such as the 16S rRNA gene or other conserved regions like the V3–V4 region are used to identify bacterial taxa (Doan et al., 2016; Dong et al., 2011; Graham et al., 2007; Lee et al., 2012a; Shin et al., 2016). This review, describes the microbes associated with the eye, both by the conventional and NGS methods in an attempt to highlight their diversity and function. It also attempts to establish a connection between ocular bacteria, antimicrobial resistance and biofilm formation.

2. Culturable ocular bacteria

Bacteria associated with the eye (ocular surface) are far fewer in numbers than those associated with other niches of the human body, this is probably because of the presence of potent antimicrobial factors in the tear film (McDermott, 2013). and the continuous up and down mechanical action of the lids (Shovlin et al., 2013). Despite these mechanisms, many microorganisms do survive on the ocular surface, as discovered by Keilty (1930) who first cultivated haemolytic *Staphylococcus* and other organisms from the conjunctival swabs of normal subjects. Subsequently, *Staphylococcus aureus*, a predominant aerobic pathogen as well as *Propionibacterium acnes* and *Peptostreptococcus* spp., which are predominant anaerobic bacteria were identified in the conjunctivae of diseased eyes (Perkins et al., 1975). The realisation that these microbes associated with the eye have health implications was an impetus for future studies on ocular bacteria; these studies led to the identification of several gram-positive (Staphylococcus epidermidis,

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Corynebacterium spp., coagulase-negative *Staphylococcus* spp., *Streptococcus* spp., *Micrococcus* spp. and *Propionibacterium acnes*) and gram-negative bacteria (*Escherichia coli*, *Neisseria* spp., *Pseudomonas aeruginosa* and *H. influenzae*) (Sharma, 2012) on the eye. Additionally, the cultured organisms recovered from conjunctival and lid swabs were significantly more in number than those obtained from the tear film (Willcox, 2013). Fungi were also identified in some of these studies (*Fusarium solani, Cladosporium sphaerospermum, Acremonium implicatum, Candida albicans* and *Aspergillus fumigatus*) (Ramage et al., 2009, 2011; Shivaji et al., 2019).

3. Identification of ocular bacteria by culture–independent methods

Abelson (2016) in a brief review of the ocular microbiome concluded that, "the placid ocular surface is actually teeming with life that might hold the secret to treating ophthalmic diseases". A resident commensal bacterial microbiome has been described for the ocular surface using NGS (Doan et al., 2016; Dong et al., 2011a; Graham et al., 2007; Lee et al., 2012; Shin et al., 2016). NGS also detected an abundance of bacteria that had previously not been identified from normal eyes, including *Rhodococcus erythropolis*, *Erwinia* spp., *Pseudomonas* spp. and *Klebsiella* spp. (Dong et al., 2011). Recent ocular surface studies indicated that bacteria affiliated to the phyla Firmicutes, Actinobacteria and Proteobacteria are the most abundant (Dong et al., 2011; Huang et al., 2016; Ozkan et al., 2017; Zhou et al., 2014). Furthermore, at the genera level, the numbers of bacteria reported from the human conjunctivae varied from a minimum of 59 genera (Dong et al., 2011) to a maximum of 600 genera (Zhou et al., 2014) and several of these genera were common in all these microbiomes and are therefore referred to as the "core genera" (Hamady and Knight, 2009). Huang et al. (2016) reported 10 core genera in all the healthy human conjunctival microbiomes with >1% mean abundance; these are *Corynebacterium* spp., *Pseudomonas* spp., *Staphylococcus* spp., *Acinetobacter* spp., *Streptococcus* spp., *Millisia* spp., *Anaerococcus* spp., *Finegoldia* spp., *Simonsiella* spp., and *Veillonella* spp. Five of these core genera namely *Pseudomonas* spp., *Corynebacteria* spp., *Acinetobacter* spp., *Staphylococcus* spp., and *Streptococcus* spp., were also reported by Dong et al. (2011). In addition to these five common genera, Dong et al. (2011) also reported 7 other core genera that included *Propionibacterium* spp., *Bradyrhizobium* spp., *Brevundimonas* spp., *Aquabacterium* spp., *Sphingomonas* spp., *Simonsiella* spp., and *Methylobacterium* spp. in the conjunctiva. In contrast to these studies, Zhou et al. (2014) identified *Corynebacterium* as the only core genus in the conjunctiva whereas Ozkan et al. (2017) did not identify any core genera. However, they did demonstrate that the ocular surface microbiome was relatively unchanged over a 3-month period and identified genera with >1% abundance in only approximately 80% of the conjunctival microbiomes; these were assigned as a "putative core", thus challenging the idea of a core microbiome. Furthermore, in various eye diseases, including dry eye disease (DED), blepharitis, and trachomatous disease (Graham et al., 2007; Lee et al., 2012a) the ocular surface microbiome varied in composition and it was not evident whether the ocular surface microbiome plays a role in the etiology of these diseases or whether these diseases alter the microbiome (Zilliox et al., 2020). These observed differences in the core genera of the conjunctival microbiomes may be attributed to variations between individuals, in sample collection methods, in specific region of the conjunctiva sampled (Huang et al., 2016), due to the continuous exposure of the ocular surface, to the external environment and to the use of tetracaine during sample collection (Dong et al., 2011; Ozkan et al., 2018; Shin et al., 2016). Although, NGS is very sensitive in identifying bacteria, the technique cannot confirm if the detected species is a viable resident of the ocular surface or not (Zegans and Van Gelder, 2014). Although the bacteria detected in the eye may simply be contaminants from the environment and (or) the regions surrounding the eye they may be considered "colonizers". Thus, a proof that a bacterium is a living resident of the ocular surface was lacking in most of these studies. In this context, it may be mentioned that *Corynebacterium mastitidis*, from the conjunctival tissue of C57BL/6 mice, which were housed in two different institutional animal facilities, was first proved to be a bona fide commensal on the ocular surface (St. Leger and Caspi, 2018). It was also proved that *C. mastitidis* is also a true commensal since it could not be acquired by commercial mice by co-housing them with the institutional mice. Furthermore, *C. mastitidis* possessed all the characteristics of a commensal, such as a better ability to colonize the ocular surface compared to other bacteria and being able to elicit the production of IL-17 which protects the ocular surface from infection (St Leger et al., 2017). In a recent study, analysis of NGS data demonstrated, that in conjunctival swabs of healthy controls, fungi affiliated to Ascomycota and Basidiomycota were the two dominant phyla (80% of the abundance) and that Zygomycota was present in as a minority population (mean abundance, 0.05%). Further, at the genera level, 80 distinct genera were detected in conjunctival swabs of healthy individuals with 11 genera being present in majority of the eyes; these were namely *Aspergillus* spp., *Setosphaeria* spp., *Malassezia* spp., *Haematonectria* spp., *Candida* spp., *Emericella* spp., *Penicillium* spp., *Fusarium* spp., *Cladosporium* spp, *Choiromyces* spp. and *Cochliobolus* spp. present in majority of the eyes (Prashanthi et al., 2019; Shivaji et al., 2019). BRiSK (biome representational in silico karyotyping), a culture independent approach, which characterizes all the DNA present in a sample using a Type IIB restriction enzyme (BsaXI) followed by sequencing (Doan et al., 2016) reported the presence of the Torque teno virus (TTV), Multiple Sclerosis–associated retrovirus (MSRV) and Human Endogenous Retrovirus K (HERV-K) in the conjunctiva. In addition, a few samples also contained the Merkel Cell Polyomavirus (MCV), human papillomavirus (HPV) and Abelson murine leukemia virus; howeve the significance of their presence was unclear (Spurgeon et al., 2015). MSRV and HERV-K are endogenous to human beings and may have originated from the host DNA. However, TTV on the ocular surface has been known to be associated with seasonal hyperacute panuveitis and with culture-negative endophthalmitis (Lee et al., 2015; Smits et al., 2012). Quantitative PCR also confirmed the presence of TTV in healthy conjunctiva and its presence was not to be influenced by gender and age.

4. Changes in the ocular microbiota changes in the diseased eye

This review will focus on bacterial taxa that have been identified as the etiological agents of ocular diseases and also highlight changes in the bacterial microbiome changes in the diseased eye wherever the data is available. Several ocular diseases and their associated microbiota are described below.

4.1. Keratitis

Keratitis is an inflammatory disease of the cornea. The infectious agents causing keratitis could be bacteria, viruses, fungi and parasites. Non-infectious keratitis could be caused by a minor injury to the eye, wearing contact lenses or the presence of a foreign body in the eye. Keratitis could lead to serious complications and loss of vision if neglected and left untreated (Green et al., 2007; McClintic et al., 2014). In India, infectious keratitis can be caused by bacteria (32.77%), fungi (34.4%), *Acanthamoeba* (1.04%) or both fungi and bacteria (2.39%) (Bharathi et al., 2007). Bacteria cultured from swabs of individuals with bacterial keratitis include several gram-negative and gram-positive bacteria with a preponderance of *P. aeruginosa*, *S. epidermidis*, *Staphylococcus* spp., *S. aureus*, *Streptococcus pneumoniae, Propionibacterium* spp., *P. acnes, Bradyrhizobium* spp. and several other species as listed in Table 1 (Bourcier et al., 2003; Egrilmez and Yildirim-Theveny, 2020; Neumann and Sjostrand, 1993; Ormerod et al., 1987; Turner and Stinson, 1965; Van Tyne et al., 2016; Zhai et al., 2018). Al-Najjar et al. (2019) examined the corneas of bacterial keratitis patients and observed that it was dominated by pathogens like *P. aeruginosa* and *Staphylococcus*

Table 1
Bacteria associated with an human ocular disease.

Ocular disease	Disease causing bacteria	Reference
Keratitis	*Actinomyces* spp., *Corynebacterium* spp., *Escherichia coli*, *Moraxella* spp., *Mycobacterium* spp., *Nocardia* spp., *Propionibacterium acne*, *Proteus* spp., *Pseudomonas aeruginosa*, *Serratia* spp., *Staphylococcus epidermidis*, *Staphylococcus* spp., *Staphylococcus aureus*, *Stenotrophomonas maltophilia*, *Klebsiella* spp. and*Streptococcus pneumoniae*.	Bourcier et al. (2003); Egrilmez and Yildirim-Theveny (2020); Neumann and Sjostrand (1993); Ormerod et al. (1987); Turner and Stinson (1965); Van Tyne et al. (2016); Zhai et al. (2018).
Conjunc-tivitis	*Chlamydia trachomatis*, *Haemophilus influenzae*, *Klebsiella* spp., *Staphylococcus aureus, Neisseria gonorrhoeae*, *Staphylococcus epidermidis*, *Moraxella catarrhalis*, *Escherichia coli* and *Streptococcus pneumonia*.	Alfonso et al. (2015); Ghosh et al. (1995); Quinet et al. (2010); Rietveld et al. (2004).
Blepharitis	*Streptophyta* spp., *Corynebacterium* spp., *Enhydrobacter* spp., *Bacillus oleronius, Staphylococcus aureus* and *Streptococcus pyogenes.*	Lee et al. (2012b); Szkaradkiewicz et al. (2012).
Scleritis	*Pseudomonas aeruginosa*, *Corynebacterium diptheriae*, *Mycobacterium* spp., *Mycobacterium tuberculosis*, *Staphylococcus pneumoniae*, *Nocardia* spp., *Stenotrophomonas maltophilia* and *Streptococcus* spp.	Bloomfield et al. (1976); Jain et al. (2009); Reddy et al. (2015).
Endoph-thalmitis	*Enterococcus faecalis*, *Escherichia coli, Neisseria meningitidis, Klebsiella* spp., *Micrococcus* spp., *Staphylococcus aureus*, *Moraxella* spp., *Staphylococcus* pyogenes, *Serratia* spp., *Streptococcus viridians*, *Streptococcus* spp., *Propionibacterium acnes*, *Bacillus cereus* and *Staphylococcus epidermidis*. *Pseudomonas* spp., *Proteus* spp., *Haemophillus influenzae*, *Klebsiella* spp., and coliform bacteria.	Bodé et al. (1985); Callegan et al. (2002); Driebe et al. (1986); Maitray et al. (2019); Tseng et al. (1996); Wong et al. (2000).
Orbital cellulitis	*Staphylococcus* spp., *Staphylococcus aureus*, *Veillonella*spp.,*Streptococcus* spp., *Haemophilus influenzae*, *Propionibacterium acne*, *Pseudomonas* spp., *Escherichia coli, Fusobacterium lentum*, *Fusobacterium necrophorum*, *Peptostreptococcus* spp., *Eikenella corrodens*, *Arcanobacterium haemolyticum*, *Moraxella catarrhalis*, *Bacteroides* spp., *Prevotella* spp. and *Fusobacterium necrophorum*.	Malik et al. (2004); McKinley et al. (2007); Partamian et al. (1983); Sharma et al. (2013); Weiss et al. (1983); Youssef et al. (2008).
Dacryo-cystitis	*Citrobacter* spp., *Entrobacter* spp., *Escherichia coli*, *Haemophilus influenzae, P. vulgaris, Pseudomonas* spp., *Staphylococcus* spp., *Staphylococcus aureus*, *Staphylococcus epidermidis, P. mirabilis, Klebsiella pneumoniae*, *Streptococcus pneumoniae* and *Streptococcus pyogenes*, *Propionibacterium acne* and *Peptostreptococcus* spp., and *Pseudomonas aeruginosa*.	Assefa et al. (2015); Briscoe et al. (2005); Chaudhry et al. (2005); Eslami et al. (2018); Kebede et al. (2010); Mills et al. (2007).
Retinitis	*Treponema pallidum, Bartonella henselea*, *Mycobacterium turberculosis*, *Borrelia burgdorferi*; *Toxoplasma gondii.*	Chi et al. (2012); Gupta et al. (2007); Hadfield and Guy (2017); Moradi et al. (2015); Sathiamoorthi and Smith (2016).
Post-fever retinitis	Rickettsia spp., *Salmonella typhi*	Mahendradas et al. (2020); Prabhushanker et al. (2017); Relhan et al. (2014).
Uveitis	*Mycobacterium tuberculosis*, *Chlamydia trachomatis*, *Treponema pallidum* spp. and *Borrelia burgdorferi.*	Barisani-Asenbauer et al. (2012).
Dry eye disease	Coagulase negative *Staphylococci*, *Cornyebacterium* spp., *Propionibacterium* spp., *Rhodococcus erythropolis*, *Klebsiella oxytoca* and *Erwinia* spp.	Albeitz and Lenton (2006); Graham et al. (2007); Kohanim et al. (2016).
Meibomian gland dysfunction	*Staphylococcus* spp., *Staphylococcus lugdunesis*, *S. aureus*, *Lysinibacillus* spp., Microbacteriaceae, *Bacillus* spp., *Moraxella osloensis*, Xanthomonadaceae, *Corynebacterium* spp., *Paenibacillus* spp., *S. aureus*, *Staphylococcus lugdunesis*, *Corynebacterium macginleyi*, *Acinetobacter calcoaceticus*, *Haemophilus influenzae*, *Actinomyces* spp., *Stenotrophomonas maltophilia*, *Pantoea agglomerans*, *Cladosporium* spp.	Jiang et al. (2018); Watters et al. (2016)
Chronic Stevens-Johnson syndrome	*Pseudomonas* spp., *Staphylococcus* spp., *Staphylococcus aureus*, *Streptococcus* spp., *Staphylococcus epidermidis* and *Acinetobacter* spp.	Kittipibul et al. (2020); Moeller et al. (2005); Sotozono et al. (2002).
Sjorgens syndrome	Firmicutes, Actinobacteria, Proteobacteria, Bacteroidetes	de Paiva et al. (2016).
Lax eyelid syndrome	*Corynebacterium* spp., *Hydrotalea* spp., *Ralstonia* spp., *Clostridium_XI*, *Staphylococcus* spp., and *Rothia* spp.	Zilliox et al. (2020).
Ocular graft-vs host disease	*Staphylococcus* spp., *alpha-hemo Streptococcus* spp., *Corynebacterium* spp., *Propionibacterium acnes*, *Haemophilus influenza*, *Enterobacteriaceae*, *Selenomonas* spp., *Pasteurellaceae*, *Lactobacillus/ Streptococcus* spp.	Zilliox et al. (2020).

spp. In a recent study, Ge et al. (2019) used a metagenomic approach to analyse the bacterial communities on the ocular surface of patients with FK and observed that the ocular surface of the healthy eyes were dominated by five genera *Pseudomonas* spp., *Corynebacterium* spp., *Staphylococcus* spp., *Thermus* spp., and *Achromobacter* spp. In contrast, in the corneal ulcer from affected eyes the dominant genera included *Pseudomonas* spp., *Psychrobacter* spp., *Achromobacter* spp., *Acinetobacter* spp., *Thermus* spp., and *Caulobacter* spp. Collectively, the FK patients displayed a disrupted conjunctival microbiome, and these alterations were implicated in the pathogenesis of FK (Ge et al., 2019).

4.2. Conjunctivitis

Conjunctivitis is also known as pink eye and is a consequence of an infection or allergy that only affects the conjunctiva. The eye is usually swollen and, may exhibit a sticky discharge, and the white of the eye is inflamed and pink in colour. Conjunctivitis, may affect one or both the eyes, exhibit a sticky discharge and could be contagious (Azari and Barney, 2013). The prevalence of conjunctivitis is 12.22% (Kahola et al., 2019). The etiological agent could be either a bacterium or a virus (Azari and Barney, 2013). *E. coli* was identified as the most common pathogen causing neonatal conjunctivitis (Ghosh et al., 1995; Quinet et al., 2010). Other bacteria that could cause conjunctivitis include several gram-negative and gram-positive bacteria including *Chlamydia trachomatis, H. influenzae, Neisseria gonorrhoeae* etc. (Table 1). Mohammed et al. (2020) used polymerase chain reaction (PCR) and 16S ribosomal DNA sequencing to identify Firmicutes (35%), Actinobacteria (31%), and Proteobacteria (26%) as the bacterial community in conjunctival sacs of eyes with infectious conjunctivitis; these communities had seven common genera namely *Streptococcus* spp., (10%), Cutibacterium spp., (10%), Staphylococcus spp., (7%), Nocardioides spp., (7%), Corynebacterium spp., (5%), Anoxybacillus spp., (5%), and Acinetobacter spp., (5%). A statistically significant difference was observed in cases with bilateral versus unilateral conjunctivitis and for cases using antibiotics.

4.3. Blepharitis

Blepharitis is an inflammatory condition of the eyelid margin (Putnam, 2016) that presents as conjunctival hyperaemia, discomfort and epiphora which eventually leads to vision loss and light sensitivity (Lemp and Nichols, 2009). In USA, 37%–47% of patients with eye diseases have Blepharitis (Lemp and Nichols, 2009) whereas in India only 12.33% of the eye diseases are caused by this condition (Lipa and Banu, 2016). Etiological factors for belpharitis include: bacterial, viral and parasitic infections, immunologic conditions, Dermatoses, benign and malignant eyelid tumors and trauma. Several bacteria have been cultured from the ocular swab samples of blepharitis patients: these include *Streptophyta* spp.*, Corynebacterium* spp.*, Enhydrobacter* spp.*, Bacillus oleronius, Staphylococcus aureus* and *Streptococcus pyogenes* (Table 1) (Lee et al., 2012b; Szkaradkiewicz et al., 2012). NGS analysis indicated that ocular microbial communities of eyelash and tear samples of individuals with blepharitis had higher relative abundances of *Staphylococcus* spp., *Streptophyta* spp., *Corynebacterium* spp., and *Enhydrobacter* spp. and lower relative abundances of *Propionibacterium* spp. Higher relative abundances of *Streptophyta* spp., *Corynebacterium* spp., and *Enhydrobacter* spp. In blepharitis subjects may indicate the presence of pollen, dust, and soil particles in the eye (Lee et al., 2012a).

4.4. Scleritis

Scleritis is a chronic, painful inflammation of the sclera and (or) adjacent tissues and could lead to vision impairment. Scleritis could be idiopathic, autoimmune or infectious (Jain et al., 2009; Pradhan and Jacob, 2013). Roughly 7–8% of all scleritis cases in adults are caused by infectious agents; and herpes zoster virus and Tuberculosis have been implicated as the most common causes of infectious scleritis (Jabs et al., 2000). Scleral inflammation is rare in children, but when it occurs, the symptoms include decreased vision and moderate to severe eye pain (Azari and Barney, 2013). The etiopathogenesis of tuberculosis scleritis is believed to be due to a direct invasion of the eye by *M. tuberculosis* bacteria or as a result of antibody-mediated inflammatory reaction against *M. tuberculosis* bacilli proteins (Bloomfield et al., 1976). In addition, the other bacteria associated with scleritis include *P. aeruginosa*, *Corynebacterium diptheriae*, *Mycobacterium* spp., *Staphylococcus pneumoniae*, *Nocardia* spp.*, Stenotrophomonas maltophilia* and *Streptococcus* spp. (Table 1) (Jain et al., 2009; Reddy et al., 2015). The microbiome of scleritis patients are yet to be described.

4.5. Orbital cellulitis

Orbital cellulitis is an inflammation of the eye behind the orbital septum, a membrane that is located in front of the orbit. The major causes of orbital cellulitis are sinusitis (58%), lid or face infection (28%), foreign bodies (11%), hematogenous causes (4%) and odontogenic causes (2–5%). The symptoms include pain, edema, chemosis, decreased ocular motility and proptosis (Chaudhry et al., 2012). Orbital cellulitis is of different types depending on the region affected and includes pre-septal cellulitis (72.2%), orbital cellulitis (22.2%) and abscess (5.6%). Orbital cellulitis appears to be more prevalent in children than in adults (Al-Madani et al., 2013; Nageswaran et al., 2006) *Staphylococcus* spp., *S. aureus*, *Streptococcus* spp. are the most common causative organisms in adults whereas *H. influenzae* is the most common cause in children. Less common causal organisms are *Pseudomonas* spp., *E. coli*, *Peptostreptococcus* spp., *Eikenella corrodens*, *Arcanobacterium haemolyticum*, and *Moraxella catarrhalis*. Anaerobes are less frequently associated with orbital cellulitis and include *Peptostreptococcus* spp., *Bacteroides* spp. and *Fusobacterium* necrophorum (Malik et al., 2004; Partamian et al., 1983; Sharma et al., 2013; Youssef et al., 2008) (Table 1).

4.6. Endophthalmitis

Endophthalamitis is an ocular disease caused by exogenous bacteria or fungi that gain access to the fluids or tissues inside the eyeball. Endogenous endophthalmitis normally occurs following eye surgeries; exogenous endophthalmitis occurs due to eye injuries (Durand, 2013). In endogenous endopthalmitis infection that occurs in other parts of the body like urinary tract or blood may spread to the eye (Chee and Jap, 2001; Connell et al., 2011; Durand, 2013; Ramakrishnan et al., 2009). Symptoms of acute endophthalmitis occur within a few days following surgery or trauma whereas in chronic endophthalmitis symptoms occur after a prolonged period of infection. The common symptoms of endophthalmitis include eye pain, white or yellow pus or discharge, red eyes, swollen eyelids and visual impairment (Durand, 2013). The incidence of endophthalmitis in India is between 0.04 and 0.15% (Lalitha et al., 2017); over the years a significant decline in the incidence of postoperative endopthalmitis has occurred due to advances in surgery, instrumentation, sterility and prophylactic antibiotic usage (Sunaric-Megevand and Pounaras, 1997). The incidence of endopthalmitis used to vary depending on the surgical procedure. For instance after extracapsular lens extraction or phacoemulsification and intraocular lens implantation it ranged between 0.07 and 0.12%; after secondary IOL implantation, this incidence was higher (0.3%). The incidence of endophthalmitis post-cataract surgery, postintravitreal injection, post-vitrectomy, post-trabeculectomy and postkeratoprostheses could range from a minimum of 0.01–0.3% (postcatarct surgery) to a maximum of 0–12.5% (postkeratoprostheses) (Levison et al., 2014). Endophthalmitis immediately following trabeculectomy is rare (0.6%) (Mandelbaum et al., 1985); however this incidence increases after the use of the antimetabolites 5-fluorouracil (5–8%) (D'Amico et al., 1985) and mitomycin C (MMC) (2.7–3%) (Wood, 1996). In most cases the patient's own flora that colonize the ocular surface of normal eyes such as *S. epidermidis*, *S. aureus* and various *Streptococcus* spp., were identified in endophthalmitis patients (Bannerman et al., 1997; Speaker et al., 1991).

Organisms causing endophthalmitis could also be introduced by contamination of the lens with microbes from the ocular surface, air of the operating room (Vafidis et al., 1984) and contaminated irrigation solutions (Egerer and Sehorst, 1976). Gram positive bacteria constitute 75–90% of all endophthalmitis infections and the predominant species causing this condition are *S. epidermidis*, *S. aureus*, *Streptococcus viridans*, *Streptococcus* spp., *Propionibacterium acnes* and *Bacillus cereus.* Gram-negative bacteria constituted 9–10% of endophthalmitis infections and included *Pseudomonas* spp., *Proteus* spp., *H. influenzae*, *Klebsiella* spp., and coliform bacteria (Bodé et al., 1985; Callegan et al., 2002; Driebe et al., 1986; Maitray et al., 2019; Tseng et al., 1996; Wong et al., 2000) (Table 1). Callegan et al. (2002) suggested a model for the occurrence of endophthalmitis based on the ability of the bacteria to induce an immune reaction. The sequence of events in endophthalmitis, include entry of the bacterium in to the eye, replication, production of toxins, stimulation of intraocular inflammation and finally inhibition of retinal function by disruption of retinal architecture and death of non-regenerating retinal photoreceptor cells. A few studies have also analysed the microbiome of ocular fluids from endophthalmitis patients (Deshmukh et al., 2019; Kirstahler et al., 2018). Microbiome analysis of vitreous from patients who developed endophthalmitis following cataract surgery or intravitreal injections predominantly included *S. epidermidis*, *Enterococcus faecalis*, *Serratia marcescens*, *Paenibacillus* spp., *Staphylococcus hominis*, *E. coli*, *Moraxella catarrhalis*, *Micrococcus luteus*, *Commamonas testosteronii* and *Caulobacter* spp. (Kirstahler et al., 2018). BRiSK analysis of the vitreous samples from endophthalmitis patients also confirmed the presence of *Staphylococcus* spp., *Streptococcus* spp., *Moraxella* spp. and *Pseudomonas* spp. (Lee et al., 2015). Furthermore, Deshmukh et al. (2019) used NGS to confirm the presence of *Streptococcus* spp., *Staphylococcus* spp., *Pseudomonas* spp. in vitreous biopsies from patients with endophthalmitis; the same study has also reported the presence of *Gemella* spp., *Haemophilus* spp. and *Acientobacter* spp. In these samples.Taken together, it appears that the microbiota in vitreous samples of endophthalmitis patients could provide leads relevant to infectious disease management.

4.7. Uveitis

Uveitis is an inflammation of the iris, ciliary body and choroid which together constitute the uveal tract; some or all the regions of the tract could be affected. Infection, trauma, and systemic diseases are the common causes of uveitis; which can also occur due to autoimmunity. Symptoms of uveitis include photophobia, pain and blurred vision and the symptoms are usually bilateral (Azari and Barney, 2013).

The prevalence of uveitis varies geographically and is around 38 per 100,000 in France, 200 per 100,000 in the US, and 730 per 100,000 in India (Sangwan, 2010). Infectious uveitis could also be caused by viruses, bacteria, fungi or parasites (Table 1) (Lin, 2015). Bacteria identified from the eyes of uveitis patients include *Mycobacterium tuberculosis*, *Chlamydia trachomatis, Treponema pallidum* and *Borrelia burgdorferi* (Barisani-Asenbauer et al., 2012). The bacterial microbiomes of the ocular surfaces and posterior chambers of uveitis patients are as yet unexplored; however several studies have provided some evidence that auto-immune and idiopathic uveitis are associated with changes in the gut bacterial microbiome (Fu et al., 2020; Horai and Caspi, 2019; Jayasudha et al., 2019; Kalyana Chakravarthy et al., 2018).

4.8. Retinitis

In Retinitis, the retina is inflamed and could be permanently damaged leading to blindness. Retinitis is also commonly referred to as retinitis pigmentosa and has a prevalence of one in every 2500–7000 people. Retinitis is one of the leading causes of blindness in patients aged 20–60 years. Several etiological agents including viruses, bacteria, ticks, parasites and fungi can cause retinitis (Williamson and Haynes, 2018). The condition manifests commonly in immune-compromised patients (Gupta et al., 2013). Viral agents causing retinitis include Varicella zoster virus (VZV), herpes simplex virus (HSV) and cytomegalovirus (CMV) (Guex-crosier et al., 1997). Bacterial causes of retinitis include *Treponema pallidum (*Moradi et al., 2015*), Bartonella henselea* (Chi et al., 2012), *M. tuberculosis* (Gupta et al., 2007), *Borrelia burgdorferi* (Sathiamoorthi and Smith, 2016), *Toxoplasma gondii* (Hadfield and Guy, 2017) etc. (Table 1). *Candida* spp. and *Aspergillus* spp. are the common causes of fungal retinitis (Invernizzi et al., 2018; Silva et al., 2015). The bacterial microbiome of individuals with retinitis has not been studied as yet although the changes in the microbiome of individuals with post-fever retinitis have been reported (see section 4.10).

4.9. Post-fever retinitis

Post-fever retinitis (PFR) is an infectious ocular disease caused by bacteria, fungi and viruses in tropical countries (Arunasri et al., 2020a, b; Kawali et al., 2019; Mahendradas et al., 2020; Vishwanath et al., 2014). Ocular manifestations of the condition are visible a few days or weeks after the onset of febrile illness and very often the systemic signs are no longer obvious. In rickettsial retinitis it is not clear whether retinitis occurs due to direct invasion of the retinal tissue by the organism or if the condition is the result of an immune driven process (Mendívil and Cuartero, 1998). PFR caused by *Salmonella typhi* infection following typhoid fever has been reported and the ocular signs included focal or multifocal retinitis (Prabhushanker et al., 2017; Relhan et al., 2014). Viruses associated with PFR include: dengue (Flavivirus serotypes DENV-1, DENV-2, DENV-3 and DENV-4), West Nile Chikungunya and Zika virus (Zhao et al., 2017). In a recent study, ocular bacterial microbiomes were generated from vitreous samples collected from normal individuals and those with post fever retinitis. A total of 18 genera were found to have differential abundance in the vitreous microbiomes of the two groups; of these, the predominant genera were *Clostridium* spp., *Enterobacter* spp., *Acinetobacter* spp., *Klebsiella* spp. and *Lachnoclostridium* spp. (Arunasri et al., 2020a). The two genera that were more abundant in PFR individuals included *Tanerrella* spp. and *Pimetobacter* spp. These genera were inferred to have antimicrobial, anti-inflammatory, and probiotic function (Arunasri et al., 2020a). The same authors also demonstrated significant changes occur in the mycobiome of vitreous samples of post fever retinitis patients thus implying that dysbiotic changes in the fungal vitreous microbiome are associated with PFR (Arunasri et al., 2020b).

4.10. Dacryocystitis

Dacryosystitis is an infection or inflammation of the nasolacrimal sac leading to blockage of the nasolacrimal duct. Symptoms include pain, redness and swelling of lower eyelid. A cross sectional study showed that the prevalence of dacryocystitis was 0.67% (Lipa and Banu, 2016). Etiological factors include bacteria, fungi and viruses (Chaudhary et al., 2010; Lipa and Banu, 2016). The common bacteria associated with dacyosystitis include *E. coli, H. influenzae, Proteus vulgaris, Staphylococcus aureus, Staphylococcus epidermidis, Streptococcus pneumoniae, Streptococcus pyogenes etc.* (Table 1). In chronic dacryocystitis, the majority of the microorganisms were gram-positive bacteria, with a predominance of *Staphylococcus* spp. and the gram-negative bacteria associated with the condition included *H. influenzae*. The most frequently isolated anaerobic microorganisms from dacryocystitis patients were *Propionibacterium acne* and *Peptostreptococcus* spp. (Chaudhry et al., 2005). Subsequently, it was confirmed that in chronic dacryocystitis patients the predominant ocular bacteria were the coagulase-negative staphylococci, *S. aureus* and *Streptococcus pneumoniae* whereas in acute dacryocystitis *S. aureus* and *P. aeruginosa* were dominant (Bharathi et al., 2008).

4.11. Meibomian gland dysfunction (MGD)

The Meibomian glands (MG) secrete oil called meibum which along with water and mucus form the composite tears. Meibum in the tears prevents the water from evaporating. However, under certain conditions, the duct of the MG is blocked or altered and as a consequence the secretory function of the gland is affected (Groden et al., 1991). This means that less meibum reaches the eye surface which in turn causes evaporative dry eye disease (DED). It was observed that the gram-positive bacteria *Staphylococcus* spp., *Lysinibacillus* spp., *Microbacteriaceae* and *Bacillus* spp. as well as the gram-negative bacteria *Moraxella osloensis* and *Xanthomonadaceae* were dominant in the normal eyes and only present in moderate numbers in eyes with MGD. However, in the severe MGD cases, in addition to the above bacteria, other gram-positive bacteria including *Corynebacterium* spp. *C. macginleyi* and *Paenibacillus* spp. were also identified; this indicates that the compositions of the cultivable microbiota in the normal and the mild MGD eyes were simple, whereas they were more abundant and complex in severe MGD cases (Jiang et al., 2018). However, subsequent results reported by Nattis et al. (2019) were only partly in agreement with the above study. Nattis et al. (2019) reported the presence of *Corynebacterium* spp., *Candida parapsilosis* and *S. aureus* in normal eyes; *Staphylococcus lugdunesis* and *S. aureus* in mild MGD patients; *Corynebacterium* spp., *Acinetobacter calcoaceticus* and *H. influenzae* in individuals with mild to moderate MGD; and *Actinomyces* spp., *Corynebacterium* spp., *Stenotrophomonas maltophilia*, *Pantoea agglomerans*, *Cladosporium* spp. in individuals with clinically significant MGD. A few studies also indicated similar microbiota were present in eyes with and without MGD (Watters et al., 2016). Dong et al. (2019) used gender and age-matched individuals to determine the abundances of *Staphylococcus* spp. (20.71% vs. 7.88%) and *Sphingomonas* spp. (5.73% vs. 0.79%) in MGD patients and normal people. The study found that the abundances of these two bacteria were significantly higher in the eyes of MGD patients than in those of normal people; in contrast, the abundance of *Corynebacterium* was significantly lower in the eyes of MGD patients than in those of normal people (20.22% vs. 46.43%). Therefore the study concluded that *Staphylococcus* spp., *Corynebacterium* spp., and *Sphingomonas* spp. may play roles in the pathophysiology of MGD (Dong et al., 2019). In a more recent study (Zhao et al., 2020) analysed changes in the ocular surface microbiomes of eyelid skin, conjunctiva, and meibum in healthy controls and patients afflicted with MGD. The study found that meibum from MGD eyes contained an abundance of *Campylobacter coli*, *Campylobacter jejuni* and *Enterococcus* faecium pathogens, which were almost absent in normal eyes. (Janssen et al., 2008; Kuriyan et al., 2014). These 3 pathogens *Campylobacter coli*, *Campylobacter jejuni* and *Enterococcus faecium* are known to cause eye infections (Janssen et al., 2008; Kuriyan et al., 2014). Functional annotation indicated that the microbiota of the meibum from MGD eyes displayed much stronger immune evasive virulence, than that of the normal eyes (Zhao et al., 2020). In yet another recent study, it was observed that the most abundant bacterial species in the meibum of healthy controls were *Pseudomonas acnes*, *Pseudomonas* spp., *Corynebacterium* spp. or the *Neisseriaceae* family; although the study also showed that the ocular microbiomes of elderly subjects were different from those of young subjects and the sex of the subjects had little impact on the composition of the bacterial communities. More importantly, it was also observed that the microbiomes of the meibum and conjunctival-sacs were different from those of the skin (Suzuki et al., 2020).

4.12. Dry eye disease (DED)

DED is a multifactorial chronic disease affecting 5–31% of the population. It can be caused by a decrease in tear production or increased tear evaporation resulting in an unstable tear film, ocular surface inflammation, and associated symptoms of ocular irritation (Bron et al., 2014; Helmick et al., 2008). Symptoms include ocular burning, itching, foreign body sensation, photophobia, redness, and reduced visual acuity (Kassan and Moutsopoulos, 2004). Prevalence rates of DED are greatest among women and the elderly (Brito-Zerón and Ramos-Casals, 2014). Several, studies indicated an association between DED and ocular surface infections as DED patients had a greater bacterial loads than healthy patients (Albietz and Lenton, 2006). Several studies have identified the following genera in the conjunctivae of DED patients: *Streptococcus* spp., *S. epidermidis*, coagulase-negative *Staphylococcus* spp., *Micrococcus* spp., *Corynebacterium* spp., *Propionibacterium* spp. and *Propionibacterium acne* (Graham et al., 2007; Hori et al., 2008; Suto et al., 2012; Watters et al., 2017; Zilliox et al., 2020). Furthermore, Graham et al. (2007) reported that coagulase-negative *Staphylococci*, *Cornyebacterium* spp., *Propionibacterium* spp., *Erwinia* spp., *Rhodococcus erythropolis* and *Klebsiella oxytoca* are more abundant in the eyes of DED patients than in those of normal people. However, Hori et al. (2008) found no significant difference in bacterial isolation rates from the conjunctivae of normal eyes and those of DED patients: however, DED patients had higher abundances of fluoroquinolone-resistant bacterial strains. It is not clear whether changes in ocular surface flora in DED are the result of the altered environment or if an altered flora contributes to the disease process (Akpek et al., 2019). Robert et al. (2002) found human herpes virus-6 (HHV-6) in the tear fluid of DED patients. Microbiome analsysis indicated that there are distinct differences between the ocular microbiome of DED patients and those of normal people. Willis et al. (2020) observed that the normal populations of *Staphylococcus* spp. in normal eyes are almost completely replaced by *Pseudomonas* spp. In patients with DED confirming the observations of Doan et al. (2016) as well as the results obtained by Graham et al. (2007). Less commonly and more inconsistent are the occurrences of *P. acnes*, *Pseudomonas* spp., *Bacillus* spp., *Acinetobacter* spp. and *Rhodococcus* spp. In DED eyes (Zilliox et al., 2020). Zilliox et al. (2020) suggested that *Corynebacterium* spp. may be one of the most frequently occurring organisms in DED eyes and that this may reflect the role of these bacteria in ocular surface disease. Differences arising in the diversities and community structures of microbes in normal and DED eyes over time have also been demonstrated by Willis et al. (2020).

4.13. Chronic Stevens-Johnson syndrome

Stevens - Johnson syndrome (SJS) is an abnormal immune-mediated response that manifests as exfoliation of the epidermis and mucosal tissue. Medications such as sulfonamides, allopurinol, paracetamol etc., stimulate SJS which could also be caused by systemic infections. Incidence of SJS is 1.2–6 out of 1,000,000 persons per year and the mortality rate in SJS patients is 1–5%. Ocular involvement has been reported in 25–75% of SJS patients (Jain et al., 2016) who typically exhibit conjunctivitis, corneal abrasion and pseudo-membrane formation (Albietz and Lenton, 2006). Recent studies focused on the differences in the ocular microbiomes of SJS patients and healthy subjects (Frizon et al., 2014; Moeller et al., 2005) and have found that SJS patients have a greater diversity of microbial species than the healthy subjects. The ocular microbiomes of SJS patients are dominated by gram-positive bacteria; gram-positive cocci make up 55% of the microbiome, followed by gram-negative bacilli (25%), and gram-positive bacilli (19%) (Moeller et al., 2005). In individuals with SJS (Kohanim et al., 2016), the microbiome of the inferior conjunctiva had more pathogenic bacteria such as *Enterobacter* spp., *Serratia nonliquefaciens, E. coli, Morganella morganii, Proteus mirabilis,* and *Haemophilus* spp. as well as commensals such as *Micrococcus* spp., and *Streptococcus* spp., than those of healthy subjects (Frizon et al., 2014; Venugopal et al., 2016). Overall, these studies reveal that a higher proportion of pathogenic microorganisms including *Pseudomonas* spp., *Staphylococcus* spp., *Streptococcus* spp. and *Acinetobacter* spp., exist in the ocular microbiomes of SJS patients (Kittipibul et al., 2020; Moeller et al., 2005); this was also confirmed by microbiome studies carried out by Kittipibul et al. (2020) and Moeller et al. (2005). These two studies also suggested that the higher number of pathogenic bacteria in SJS eyes probably influence chronic inflammation and opportunistic infections in SJS eyes (Kittipibul et al., 2020). This altered microbiome may also have arisen due to the host's abnormal innate immune system (Ueta and Kinoshita, 2012). Sotozono et al. (2002) reported that SJS was significantly associated with development of methicillin resistant *S. aureus* (MRSA) and methicillin-resistant *S. epidermidis* (MRSE) keratitis.

4.14. Sjorgen's syndrome (SS)

SS is a chronic autoimmune disease that primarily affects the body's moisture-producing lacrimal and salivary glands thus causing oral and ocular dryness. The other symtoms of SS include dry skin, vaginal dryness, a chronic cough, numbness in the arms and legs, feeling tired, muscle and joint pains and thyroid problems. SS could develop at any age and is more common in females (Brito-Zeron et al., 2016; Singh et al., 2016). It affects about 0.1–4% of the people in the USA. A study by de Paiva et al. (2016) shows that the microbiome samples from the inferior conjunctiva of eight normal control subjects, six Rosacea patients, and 15 SS patients have core microbiomes consisting predominantly of Firmicutes, Actinobacteria, Proteobacteria, and Bacteroidetes phyla in conjunctival samples from all three groups. However, the study could not find any significant differences in overall composition, richness, or structure of the ocular surface microbiome between the three study groups. However, the same investigators did observe that the stool and tongue microbiomes of normal subjects and SS patients were different (de Paiva et al., 2016; Mandl et al., 2017). In subsequent studies (Mandl et al., 2017; Moon et al., 2020), dysbiosis in the gut microbiome in SS patients was confirmed; SS patients showed significant dysbiosis compared to normal subjects and those with environmental dry eye syndrome. The composition of gut microbiomes of DED patients were somewhere in between those seen in SS patients and normal subjects (Moon et al., 2020).

4.15. Lax eyelid syndrome (LES)

Patients with lax eyelid syndrome (LES) typically have "floppy" eyelids and conjunctivitis. LES effects patients of all ages and weights (van den Bosch and Lemij, 1994) and is associated with obstructive sleep apnoea (Peppard et al., 2013). LES patients also have systemic elevation in inflammatory markers like matrix metalloproteinases (Sward et al., 2018). *Corynebacterium* spp. was detected in the majority of the LES eyes whereas *Hydrotalea* spp., *Ralstonia* spp., *Clostridium*XI, *Staphylococcus* spp., and *Rothia* spp. were detected in the eyes of a few LES patients. It was also observed that the α-diversity in the LES eye was significantly lower than that of the healthy eye (Zilliox et al., 2020).

4.16. Ocular graft-vs-host disease (oGVHD)

Graft-versus-host disease (GVHD) is an immune-mediated disease caused by complications in the hematopoietic stem cell transplantation and is a leading cause of morbidity and mortality (Ferrara et al., 2009; Mirza et al., 2016). Ocular GVHD (oGVHD) occurs in more than 60% of GVHD patients (Herretes et al., 2015); the most common presentations of oGVHD are keratoconjunctivitis sicca, cataract, blepharitis, ocular hypertension and filamentary keratitis (Lin and Cavanagh, 2015). Dry eye, conjunctivitis, keratoconjunctivitis sicca and punctate keratopathy (Ogawa et al., 1999; Shikari et al., 2013) are the symptoms of acute oGVHD; patients with oGVHD had increased eye discharge, red eye, tearing and dryness. In contrast, chronic oGVHD patients, often reported were eye dryness, increased eye discharge, photophobia and alacrimia (Ogawa et al., 2013; Qiu et al., 2018). Shimuzu et al. (2019) analysed the commensal microflora in the conjunctiva of oGVHD patients and found *Staphylococcus* spp., alpha-hemo *Streptococcus* spp., *Corynebacterium* spp., *Propionibacterium acnes*, aerobic gram-positive cocci, *Haemophilus influenzae* and aerobic gram-positive rods in the eyes of GVHD patients thus demonstrating that ocular surface microbes of oGVHD patients were more complex and diverse than those of healthy subjects. Zilliox et al. (2020) analysed the microbiome of the ocular surface of patients with oGVHD and observed that they had fewer sequencing reads compared to healthy subjects. In fact reads were detected only in 6 of the 14 oGVHD eyes that were analysed. In one of these eyes a predominance of *Enterobacteriaceae* spp., *Selenomonas* spp. and *Pasteurellaceae* was detected; none of which were present in significant numbers in the other eyes. In five of the 6 (83%) oGVHD eyes *Lactobacillus/-Streptococcus* spp. were detected; these bacteria were also seen in the healthy eyes, albeit at a lower relative abundance. Zilliox et al. (2020) also observed higher relative abundances of *Corynebacterium* spp. In ocular microbiome samples of DED patients than in those of oGVHD patients. The α-diversities in healthy were higher than in eyes affected by chronic SJS, oGVHD, LES and DED; this observation implies that the lack of bacterial diversity in these conditions may play a role in disease pathogenesis.

5. Treatment of ocular infectious diseases

Treatment of ocular infections requires administration of antimicrobial antibiotics. Bacterial infections of the cornea and conjunctiva may be cured by the use of antibiotic eye drops. Eye drops are either absorbed by the cornea and reach the intra-ocular tissue through the aqueous humor or absorbed by the conjunctiva, sclera and choroid to reach the vitreous cavity and the retina (Weijtens et al., 2002). Apart from topical antibiotics intraocular injections are used to deliver antibiotics into the anterior chamber or vitreous cavity to treat intraocular infections (Weng et al., 2017). In the last decade, treatment of ocular infections has been compromised by the emergence of anti-microbial resistance (AMR) in the bacteria associated with ocular infections (see the section below).

6. Anti-microbial resistance (AMR) in ocular bacteria

AMR is defined as the ability of a bacterium (microbe) to resist the killing effect of an anti-microbial drug. AMR is a global phenomenon and every year more than 0.7 million people die world-wide due to drug-resistant infectious diseases (Davies et al., 2013). The British government has estimated that by 2050 ten million people would die annually from AMR complications (ROAR, 2016). Therefore, to overcome this situation, governments world-wide have put in place various agencies for national surveillance of antibiotic usage and for educating the public on AMR.

Over the years, several ocular microbes have become resistant to various classes of topical antibiotics like polymyxin B combination drugs, macrolides, aminoglycosides and more recently fluoroquinolones (Sharma, 2011). Antimicrobial susceptibility testing indicated that ocular microbes like *Pseudomonas* spp., *Staphylococcus aureus*, *S. epidermidis*, *Micrococcus* spp., *S. pneumoniae*, *Haemophilus influenzae, Acinetobacter* spp., *Moraxella* spp. etc. which were hitherto susceptible to a particular drug have now become resistant (Bharathi et al., 2010; Kim and Toma, 2011; Lee et al., 2019; Lichtinger et al., 2012; Shalchi et al., 2011; Shimizu et al., 2013; Sugita et al., 2013). Kunimoto et al. (1999) examined *in vitro* susceptibility of 1558 corneal isolates from bacterial keratitis patients and observed that resistance of ocular *S. epidermidis*, *S. aureus*, *Staphylococcus* spp., *Corynebacterium* spp., *Streptococcus pneumoniae*, alpha-hemolytic *Streptococcus* spp. and *Pseudomonas aeruginosa* had higher levels of resistance to ciprofloxacin than those detected for these bacteria between 1992 and 1996. Isolates of *Streptococcus pneumoniae* from conjunctivitis patients were also reported resistant to gentamicin, tobramycin and polymyxin B (Block et al., 2000). Bharathi et al. (2010) analysed 4417 ocular isolates from eyelid infections such as endophthalmitis, corneal infections, blepharitis, conjunctivitis, keratitis and dacryocystitis and observed that Gram-positive cocci were susceptible to vancomycin and moxifloxacin, while Gram-negative bacteria were susceptible to gatifloxacin and amikacin. But in each case some of the isolates (10%) were resistant to these antibiotics. Several other studies have also confirmed that ocular bacteria like cogulase-negative *Staphylococcus* spp., *S. pneumoniae, S. aureus, S. epidermidis*, *Haemophilus influenza, Corynebacterium macginleyi*, *P. aeruginosa*, *Enterococcus faecalis* isolated from patients with keratitis, corneal ulcers, and endophthalmitis exhibited resistance to (at least few of the following) Tobramycin, Gentamicin, Azithromycin, Gatifloxacin, Moxifloxacin, Ciprofloxacin, Norfloxacin, Levofloxacin, Ampicillin, Cephalothin, Neomycin, Tetracyclins and Ciprofloxacin (Alexandrakis et al., 2000; Cavuoto et al., 2008; Chaudhry et al., 1999; Eguchi et al., 2008; Harper et al., 2007; Kaliamurthy et al., 2005; Kim and Toma, 2011; Mayo et al., 1986; Morrissey et al., 2004; Rau et al., 2008; Rishi et al., 2009). A retrospective review spanning 8 years (during 2007–2014) observed that ocular *Pseudomonas* spp., from patients with microbial keratitis, showed a significant increase in AMR, (Das et al., 2019). Joseph et al. (2019) further confirmed that many ocular organisms showed increased AMR, with gram-positive bacteria showing resistance to gatifloxacin, fluoroquinolones and cephalosporins and the gram-negative bacteria to ceftazidime (Joseph et al., 2019). As all these observations herald the emergence of AMR in ocular bacteria (Chalita et al., 2004), it is essential that we understand the basis of AMR in microbes.

7. Factors driving AMR in ocular bacteria

Microbial resistance to drugs may occur due to extrinsic or intrinsic factors. In the former case, humans contribute significantly to AMR due to the following reasons: (a) high consumption of broad spectrum antibiotics, (b) improper dosing of antibiotics, (c) improper use of antibiotics for veterinary and agricultural purposes (Buckner et al., 2018), (d) improper disposal of antibiotics, (e) lack of effective methods to prevent transmission of resistant bacteria in healthcare facilities and (f) use of sub-standard antibiotics. Several reviews address some or all of the above issues (Allcock et al., 2017; Castro-Sanchez et al., 2016; Kumarasamy et al., 2010; Muloi et al., 2019; Pai, 2015).

Antibiotics generally kill bacteria generally by targeting enzymes involved in cell wall biosynthesis, by inhibition of protein synthesis, inhibiting nucleic acid metabolism and DNA repair, and disrupting membrane structure (Fig. 1) (Li and Webster, 2018). Thus, the mechanisms by which bacteria become resistant to antibiotics serve to protect the above mentioned targets (Fig. 2) with strategies that include:

(i) modification of the cell wall to block antibiotics from entering the cell,
(ii) alteration of bacterial proteins/targets so that antibiotics bind to the targets with less affinity,
(iii) inactivation of the anti-microbial agent,
(iv) the active efflux of the antibiotic,
(v) acquiring AMR genes and plasmids which confer resistance and

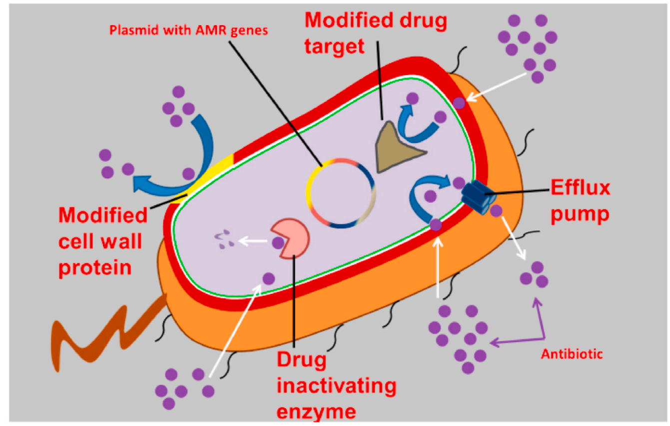

Fig. 2. Mechanisms by which bacteria become resistant to antimicrobial agents. Modified from Dantas and Sommer (2014).

(vi) by forming biofilms.

8. Biofilm formation

Biofilms are a defence mechanism by which bacterial and/or fungal cells (the viable component) sequester themselves within a protective environment, which is usually a self-secreted non living extra-polymeric substance (EPS - the abiotic component) to evade the killing effect of an antimicrobial agent (Fig. 3), the hostile environmental conditions and/or host defence mechanisms (Davies et al., 1998). The EPS maintains the structural integrity of the biofilm, anchors the biofilm to a substratum (Characklis and Marshall, 1990), facilitates cell-to-cell communication and protects the viability of cells by modulating substrate absorption, oxygen diffusion and transport of molecules within the biofilm (Branda et al., 2005; Flemming and Wingender, 2010). A characteristic feature of the bacteria involved in biofilm formation is the transition from a

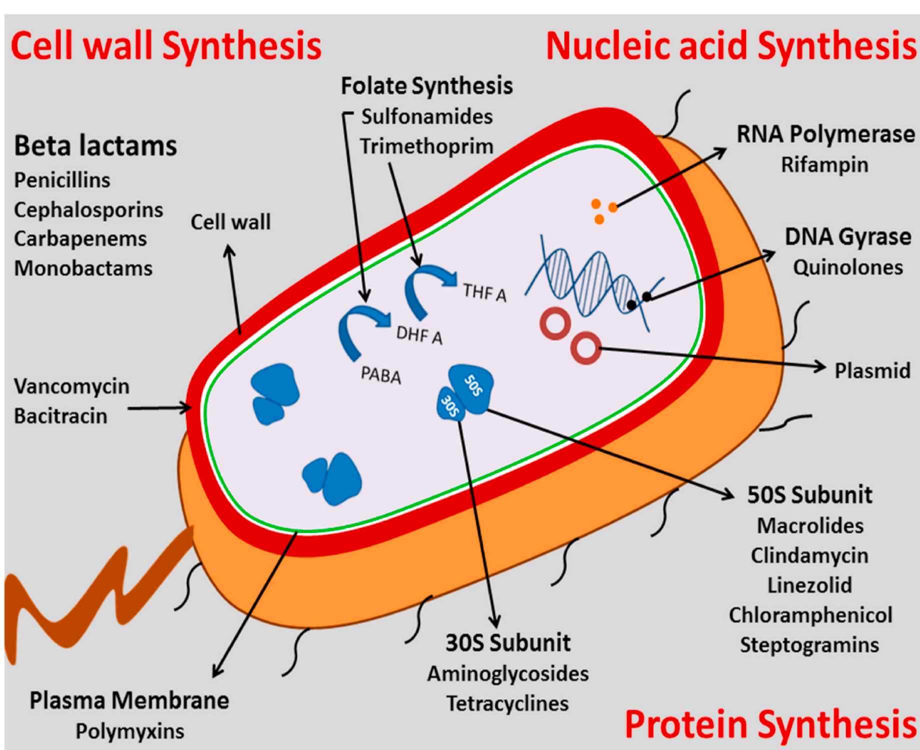

Fig. 1. The four targets of action of antimicrobial agents (protein synthesis, nucleic acid synthesis, cell wall synthesis and membrane disruption). Modified from Li and Webster (2018).

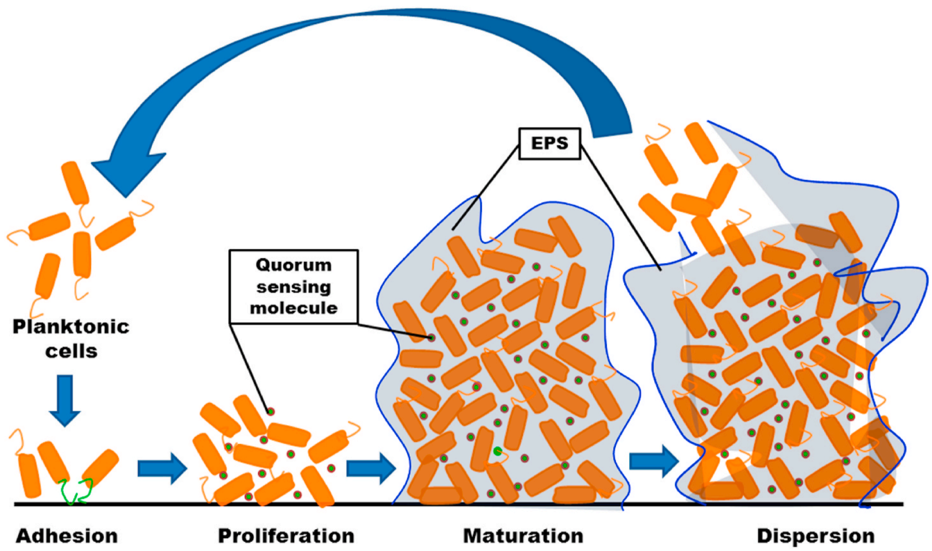

Fig. 3. Different stages (adhesion or attachment, proliferation or microcolony formation, maturation and detachment or dispersion) in the development of a biofilm in bacteria. Modified from Joo and Otto (2012).

planktonic phase to a sedentary life style on a surface (O'Toole and Kolter, 1998) to form a biofilm. This transition occurs in 4 distinct stages: adhesion, microcolony formation, maturation and dispersion (Behlau and Gilmore, 2008; Joo and Otto, 2012) (Fig. 3).

8.1. Adhesion

Biofilm formation begins when bacteria sense changes in environmental cues such as pH, osmolarity, temperature, iron levels, oxygen levels etc. which are conducive to biofilm formation (O'Toole and Kolter, 1998). Following this the cells begin to adhere to a substratum (this is first reversible and then becomes irreversible) via. physicochemical forces such as hydrophobicity, van der Waals forces and hydrophobic interactions (O'Toole et al., 2000). Additionally, bacteria like *S. epidermidis* and *S. aureus* employ several other surface proteins such as autolysin (*atl* and *atlE*) (Buttner et al., 2015; Gross et al., 2001; Ormerod et al., 1987), motility organelles such as fimbriae (Blumer et al., 2005) and pili (Deziel et al., 2001), exo-polysaccharides (Ma et al., 2009), extracellular DNA (Pakkulnan et al., 2019) and bacterial microbial surface components recognizing adhesive matrix molecules (MSCRAMMs) (Clarke and Foster, 2006; Foster and Hook, 1998) that interact with matrix proteins such as fibrinogen (Pei et al., 1999) and fibronectin (Li et al., 2012; Maxe et al., 1986; McElroy et al., 2002) to adhere to the substratum.

8.2. Microcolony formation

In stage 2, bacteria proliferate and become organised into multilayered cellular micro-colonies. This step is mediated by polysaccharides such as intercellular adhesin as in *S. epidermidis* and *S. aureus* (Mack et al., 1994, 1999; Rohde et al., 2007), specialised surface proteins like Aapas in *S. epidermidis* (Bowden et al., 2005; Schaeffer et al., 2015) and the homologous *S. aureus* surface protein G (SasG) (Bateman et al., 2005; Conrady et al., 2013). In these microcolonies the bacteria divide and organise themselves in to simple or elaborate structures (Pilchová et al., 2014).

8.3. Maturation

In stage 3, the biofilm is fully developed and appears as vertical columns or mushroom-like assemblies with distinct water channels, these water channels provide nutrients to the deeper biofilm layers and facilitate waste exchange (Jahid and Ha, 2012; O'Toole et al., 2000). The cells in the biofilm begin to secrete EPS at this stage which appears as a gelatinous material comprised of proteins, polysaccharides, nucleic acids, lipids, dead bacterial cells, and other hydrated polymeric substances whose compositon may be 85–95% water (Costerton et al., 1981). By the end of this stage the biofilm prepares for stage 4, which is the dispersal phase.

8.4. Dispersal

The dispersal phase is characterized by an active phenotypic switch involving the sensing of environmental cues and their transduction through complex regulatory networks to allow the release of individual cells and/or multicellular aggregates from the biofilm (Guilhen et al., 2017). This process is facilitated by polysaccharide-degrading enzymes such as beta-mannanase (Dow et al., 2003), chitinases (Baty et al., 2000), exopolysaccharide lyase (Allison et al., 1998) and alginate lyase (Boyd and Chakrabarty, 1994). In addition several proteases (Cui et al., 2016; Lister and Horswill, 2014; Marti et al., 2010) and nucleases (Tang et al., 2008; Tetz et al., 2009; Whitchurch et al., 2002) are also involved in biofilm dispersion. Other strategies for dispersion include cell autolysis (Ma et al., 2009), release of small peptides called phenol-soluble modulins and surfactants (like rhamnolipids) (Aleksic et al., 2017; Angelini et al., 2009; Bonnichsen et al., 2015). Passive detachment of cells from the biofilm induced by environmental changes and by chemical means or force (Kaplan, 2010) may also contribute to dispersion. It is also suggested that Quorum sensing is involved in this critical stage.

9. AMR in ocular bacteria is associated with biofilm formation

This review focuses specifically on AMR associated with biofilm formation in ocular isolates. Several ocular bacteria have been reported to form biofilm on intraocular lenses, contact lenses, suture material, lid implants, socket implants, orbit implants and scleral buckles (Zegans et al., 2002). *Pseudomonas aeruginosa*, *S. epidermidis*, *Streptococcus* spp. and *Enterobacter* spp. have been identified as components of the above biofilms (Elder et al., 1995). Katiyar et al. (2012) demonstrated that 85% of the isolates from intraocular lenses are represented by *P. aeruginosa*, *Staphylococcus aureus, S. epidermidis, Micrococcus luteus, S. marcescens, Neisseria* spp.*, Moraxella* spp.*, Bacillus* spp.*, E. coli, Proteus mirabilis*, *Enterobacter agglomerans* and *Klebsiella* spp., which were resistant to antibiotics. Furthermore, it has been demonstrated that the minimum inhibitory concentrations (MIC) of antibiotics for bacteria in biofilms are much higher than those for bacteria in the planktonic phase. For instance, in ocular bacteria like *S. aureus, S. marcescens, Klebsiella pneumoniae, P. aeruginosa, S. pyogenes, S. epidermidis, S. pneumoniae,* coagulase negative *Staphylococcus* spp. and *E. coli* the MICs of an antimicrobial agents are usually >100 fold higher for the biofilm phase as compared to that of the planktonic phase cells (Jassim et al., 2015b; Kobayakawa et al., 2005; Maciejewska et al., 2016; Pichi et al., 2014). This increase in MIC of antibiotics in the biofilm phase could be attributed to: (a) inefficient penetration of the drug into the biofilm (Gordon et al., 1988; Mah and O'Toole, 2001; Stewart, 1998), (b) inability of the drug to exerts its effect within the biofilm (Tack and Sabath, 1985; Xu et al., 2000; Zhang and Bishop, 1996), (c) transformation of the microorganisms in the biofilm into viable-but-nonculturable state (Brown et al., 1988; Li et al., 2014), (d) emergence of persister cells which are resistant to drugs (Gerdes and Maisonneuve, 2012), (e) ability to survive under nutrient and oxygen limitation conditions, (f) up-regulation of drug resistance-associated genes (Elder et al., 1995; Hoyle et al., 1992), (g) ability of EPS to limit diffusion of Aminoglycosides (Fux et al., 2005; Khan et al., 2010), (h) ability of EPS to inactivate antibiotics (Billings et al., 2015; Daddi Oubekka et al., 2012), (i) horizontal gene transfer within the biofilm community (de Kievit, 2009), which may result in acquired resistance to phagocytosis and induction of LPS modification genes (Wilton et al., 2016). AMR in ocular isolates may arise due to biofilm formation, and AMR genes (the presence of these genes was concordant with AMR in 10 ocular *E. coli* strains), presence of several virulent genes (*fimB* to *fimI*, *papB* to *papX*, etc.) and prophages (Enterobacteria phage HK97, Enterobacteria phage P1, *Escherichia* phage *D108* etc.) which were unique to ocular *E. coli* (Ranjith et al., 2017, 2020). Genes involved in biofilm formation in ocular bacteria are discussed in detail in section 11 below.

10. Quorum sensing and biofilm formation in ocular bacteria

"Quorum sensing" (QS) refers to the density-dependent control of gene expression in bacteria leading to the upregulation of several genes (>600) (Davies et al., 1998; Suga and Smith, 2003; Wagner et al., 2003) critical for an array of physiological activities which including motility, virulence, symbiosis, sporulation, conjugation and biofilm formation. In this review QS is mentioned only as a step preceding biofilm formation. In most gram-negative bacteria acyl-homoserine lactones (AHLs), functioning as signalling molecules (auto-inducer molecules). When the AHLs are synthesized in a bacterial cell and cross once the threshold concentration the AHLs diffuse out, enter nearby cells and bind to LuxR or LuxL (the response regulators) which are the cytoplasmic transcription factors. The AHL- LuxR/LuxL complex binds to DNA to drive the transcription of quorum-specific genes (Fig. 4) (Papenfort and Bassler, 2016). In contrast, in gram-positive bacteria autoinducing peptides (AIPs) such as cyclic lactone peptides or acyclic oligopeptides act as the Quorum sensing molecules (QSM). AIP are usually synthesized as a precursors and are transformed into mature AIPs before being secreted into the extracellular environment. When a threshold concentration of AIPs is attained, they molecules bind to a sensory histidine kinase on the membranes of the bacterial cells. This kinase is activated by AIP binding and undergoes autophosphorylation; it then phosphorylates a cytoplasmic response regulator which binds to DNA and drives the transcription of quorum-specific genes (Papenfort and Bassler, 2016; Rutherford and Bassler, 2012) and triggers biofilm formation. In ocular bacteria (Elder et al., 1995) QS has not been studied in detail. N-acyl--homoserine lactone (AHL) has been identified as the QSM in some ocular Gram-negative bacteria (*Aeromonas hydrophila*, *P. aeruginosa*, *Serratia marcescens* and *Serratia liquefaciens*) but not in *E. coli*, *Stenotrophomonas maltophilia, Acinetobacter* spp., *H. influenza* and *Klebsiella oxytoca* (Zhu et al., 2001, 2002). QS involving LASL and LASR or *RHL*R and *RHL*AB and the QSM AHL have been detected in ocular *P. aeruginosa* isolated from microbial keratitis patients (Zhu et al., 2002, 2004); however, these systems have not been detected in other ocular bacteria.

11. Candidate genes involved in biofilm formation in ocular bacteria

Although, several approaches have been used to identify genes involved in biofilm formation such as large-scale mutant screens (Oluyombo et al., 2019), promoter identification assays (Amores et al., 2017; Finelli et al., 2003), transcription profiling using DNA microarrays (Hancock and Klemm, 2007; Junker et al., 2007; Ren et al., 2004; Schembri et al., 2003) and proteome analyses (Sauer et al., 2002); such studies on ocular bacteria are few. An ocular strain of *E. coli* L-1216/2010, from the vitreous fluid of a patient with endophthalmitis,

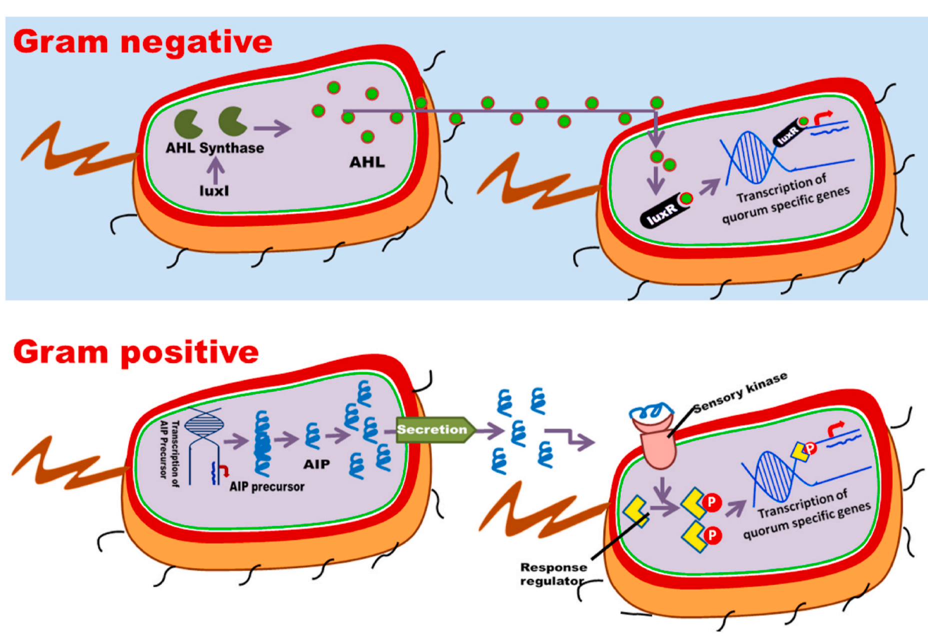

Fig. 4. Mechanism of QS in Gram-negative and Gram-positive bacteria. AHL, Acyl-homoserine lactones; LuxR, the response regulator; AIP, auto-inducing peptides.

which was resistant to eight different antibiotics and exhibited biofilm formation, was used as a model system to identify genes over-expressed in the biofilm phase compared to the planktonic phase cells (Ranjith et al., 2017). DNA microarray analysis indicated that several genes were either up- (426) or down-regulated (866) in the biofilm forming *E. coli* (L-1216/2010) cells compared to those in the non-biofilm forming *E. coli* L-1339/2013 cells. Annotation of the genes indicated that the up-regulated genes belonged to different functional categories and were likely to be required for biofilm formation and AMR. In addition, in a follow-up study the authors also demonstrated using the same ocular *E. coli* (L-1216/2010) and candidate gene mutation studies that several genes were absolutely required for biofilm formation (Ranjith et al., 2019). All the above approaches indicated that in ocular bacteria the genes required for biofilm formation could be categorised as those involved in motility, adhesion, lipopolysaccharide synthesis, biofilm architecture, drug and active transport, pathogenicity, virulence, DNA repair and stress: in addition, genes such as *ryf* coding for small RNA, *tol*A, the metabolic genes *mph*A and *mph*B, *bdc*R, *ytf*R, *mdt*O etc. (Table 2) (Ranjith et al., 2017, 2019) are involved in biofilm formation. A few biofilm genes like *atlE* and *ica* were detected in ocular coagulase-negative staphylococcus (Jassim et al., 2015a), *ica* and *icaB were found in S. haemolyticus, S. hominis, S. capitis, S. warneri, S. lugdunensis* and coagulase-negative staphylococci (Makki et al., 2011), *icaA, ica*D, *arc*A, and *opp*3AB were detected in *S. epidermidis* (Flores-Paez et al., 2015; Jena et al., 2017) and *ladS* was found in *P. aeruginosa* (Dave et al., 2020). In ocular *E. coli* it was also observed that mutations in some genes involved in biofilm development were responsible for inhibition of motility/swimming/swarming (Δ*tolA*), adhesion (Δ*tolA*), production of EPS (Δ*bdcR*, Δ*mhpA*, Δ*mhpB*, Δ*ryfA*, and Δ*tolA* mutant), inhibition of curli fimbriae and cellulose nanofibers production (Δ*bdcR* and Δ*ryfA*) (Ranjith et al., 2019).

12. Candidate genes involved in biofilm formation in non-ocular bacteria

A few genes such as *ryfA* (Whiteley et al., 2001; Bak et al., 2015), *tolA* (Vianney et al., 2005; Whiteley et al., 2001), *bdcA* (Ma et al., 2011), *ytfR* (Pinweha et al., 2018), *ymdB* (Kim et al., 2013), *csrA* (Jackson et al., 2002), *spo*T and *relA* (de la Fuente-Nunez et al., 2014) and *ydgG* (Herzberg et al., 2006) have been shown to be required for biofilm formation (by using specific-knock out gene experiments) either in *P. aeruginosa*, *E. coli* or *Burkholderia pseudomallei* (Vianney et al., 2005). Niba et al. (2007) screened 3985 single-gene deletion mutants of *E. coli* named the 'Keio collection' and identified 110 genes involved in biofilm formation; these included 40 genes coding for motility, 7 coding for Type I fimbriae, 6 curli genes, 8 LPS genes. The remaining genes coded for a variety of metabolic functions and physiological processes and included all the genes listed above. In another study Alav et al. (2018) listed 21 efflux pump genes in *E. coli,* which when knock out either reduced (12 genes) or enhanced (4 genes) biofilm formation; 5 of these genes were not evaluated for biofilm formation. Efflux pumps are known to confer AMR to bacteria.

(Levy, 2002) and several genes for these pumps (*emrY, fsr, ydeA*, acrB, acrD, acrE, emrE and tolC) were also found to be important for biofilm formation (De Kievit et al., 2001; Junker et al., 2006). The transport genes *MexCD-OprJ* and *MexAB-OprM* have also been implicated in biofilm formation (Gillis et al., 2005). Several other genes that were implicated in biofilm formation in non-ocular bacteria include genes involved in: (a) motility (*cya*A and *crp),* (b) biosynthesis of the flagella (Chilcott and Hughes, 2000), (c) Type1 fimbriae (coded by the genes *fim* A to *fim*I), (d) the *fim* like genes *fhi*A and *ECs1554* (e) curli formation (*csg*A to *csg*G, *cya, ihf*B, *lon, mlr*A, *nlp*D, *omp*R, *rpo*S, *and yie*O) (Hancock and Klemm, 2007; Niba et al., 2007; Prigent-Combaret et al., 2000; Ren et al., 2005), (f) LPS synthesis (*ipc*A, *gmh*B, *rfa*D, *rfa*E, *rfa*F, *rfa*G, *rfa*H, and *rfa*P*)* (Hancock and Klemm, 2007; Niba et al., 2007), (g) biofilm architecture such as cellulose synthesis encoding genes (*bcs*ABZC and *bcs*EFG genes) (Hancock and Klemm, 2007), (h) colanic acid synthesis (*wca*L and *wca*M) (Hancock and Klemm, 2007), (i) pathogenicity and virulence genes (*rfaH, c2418 to c2440*) (Hancock and Klemm, 2007) and (j) stress genes (*csp*G, *csp*H, *pph*A, *ibp*A, *ibp, sox*S, *hha* and *yfi*D) (Beloin et al., 2004; Hancock and Klemm, 2007; Ren et al., 2004; Schembri et al., 2003).

Table 2
Genes involved in biofilm formation in ocular *Escherichia coli* as deduced from DNA microarray analysis or by the analysis of gene knockout mutants*.

S. No.	Gene	Description
1	Motility genes	
	fhiA	encodes a novel flagellar system
	ECs1554	encodes a putative minor tail protein
2.	Adhesion genes	
	yadL, *yad*M, Z5029 and Z1651	encoding for putative fimbrial-like adhesin proteins (Reisner et al., 2003)
3.	Lipopolysaccaride genes	
	waaB, waaP, waaJ and *waaR*	encoding for lipid A moiety of LPS
	waaU and *waaY*	encoding for the phosphorylated core oligosaccharide of LPS
	waaL and *wzzB*	encoding for the O antigen of LPS
4.	Biofilm architecture	
	yhjN	encoding for cellulose synthase regulator
	wcaL and *wcaM*	encoding for colanic acid synthesis
5.	Drug resistance, transport and active transport	
	*mdl*A and *mdt*M and *mdt*O	encoding for multidrug drug efflux pumps
	*emr*B	confers resistance to cyanide *m*-chlorophenyl-hydrazone, tetrachlorosalicylanilide, organo-mercurials, nalidixic acid and thiolactomycin,
	sat	encoding for the secreted auto-transporter toxin
	ompC	encoding for secretion of extracellular proteins,
	htrE	encoding putative chaperone-usher fimbrial protein
	mhpA, mhpB,mhpF, yiaY, yajO/ydbK and *yjjN*	encoding dehydrogenases
6.	Pathogenicity and virulence genes	
	ECs3276	encoding a virulence protein populating the cluster O80301 (Baisa et al., 2013).
	sat, yjgK, *chpS*, *chpB* and *ygjN*	encoding for toxin production and secretions that governs the virulence in pathogenic bacterial cells
	wzzB and *waaL*	encoding proteins that influence antigen presentation.
	yiiX	encoding extracellular matrix remodelling
	cbrB, *cbrC*, *hisI* and mgl*B*	transport of amino-acids and other metabolites
7.	DNA repair and stress genes	
	iraD, *uvrD* and *recC*	encoding genes involved in DNA damage repair
	speB, *ybeW*, *hscC* and *djlC*	encoding genes involved in stationary phase stress
	dsmB, *glpB* and *nrdD*	encoding genes involved in anaerobic stress
	uspC	encoding a universal stress protein (Beloin et al., 2004; Hancock and Klemm, 2007; Ren et al., 2004; Schembri et al., 2003)
8.	*ryfA*	encoding a small RNA (Bak et al., 2015)
9.	*tolA*	ecoding for the inner membrane protein -cell envelope integrity
10.	*bdcA* and *bdcR*	Encoding bdcR the regulatory gene of *bdcA*
11.	*atlE*	coagulase-negative *Staphylococcus*
12.	*icaA* and *icaB*	coagulase-negative *Staphylococcus*, *S. warneri*, *Staphylococcus haemolyticus*, *S. hominis*, *S. capitis*, *S. lugdunensis*
13.	*icaA*, *icaD*, *arcA* and *opp*3AB	*Staphylococcus epidermidis*
14.	*ladS*	*Pseudomonas aeruginosa*

*Data from Dave et al. (2020); Flores-Paez et al. (2015); Jassim et al. (2015b); Jena et al. (2017); Makki et al. (2011); Ranjith et al. (2017); Ranjith et al. (2019).

13. Hacking biofilms to overcome AMR

The strategies to eradicate pathogens in the biofilm phase need to be different from those used to treat non-biofilm forming infection, as most drugs cannot penetrate the biofilm sufficiently to kill the cells within. Novel antibiofilm agents have been studied for their potential to control biofilm-related infections (Sharma et al., 2019); these are:

(i) D-amino acids which have been used to disperse biofilms in *S. aureus, P. aeruginosa, Bacillus subtilis* and *E. coli* biofilms (Hochbaum et al., 2011; Kolodkin-Gal et al., 2010; Romero et al., 2011). The mechanism of action of these agents was attributed to the ability of the D-amino acids to trigger biofilm disassembly by preventing cell surface-associated proteins to connect neighboring cells in large aggregates (Geoghegan et al., 2010).

(ii) Using a combination of an antibiotic with Tween 80 and N-acetylcysteine (NAC) to disperse biofilms (Munoz-Egea et al., 2016). In this method it was observed that amikacin, ciprofloxacin, and clarithromycin, alone and in association with N-acetylcysteine and Tween 80, dispersed rapidly growing mycobacteria biofilms.

(iii) Enzymatic degradation of the biofilm matrix as a strategy to disperse biofilms. This strategy uses enzymes like Dispersin B, DNase I, proteases etc. to disperse biofilms of several different bacteria (Sun et al., 2013; Tetz et al., 2009). Since the EPS matrix in a biofilm acts as a barrier to antimicrobial compounds, disrupting the EPS matrix could disperse mature biofilms. It is well known that enzymes endogenous to biofilm cells degrade EPS to facilitate dispersal of the biofilm cells and formation of new biofilms (Flemming et al., 2016). Dispersin B, a glycosyl hydrolase, is one such enzyme which disrupts poly-N-acetylglucosamine, the main exopolysaccharide in biofilms (Chaignon et al., 2007). This enzyme was used effectively against biofilms of *S. epidermidis*, *Burkholderia cenocepacia*, and *Achromobacter xylosoxidans* (Dobrynina et al., 2015). DNase I has also been used for the same purpose and DNase I in combination with several antibiotics (Tetz et al., 2009) disrupts the matrix material and increases the killing efficiency of antibiotics (Tetz et al., 2009; Torelli et al., 2017) in *Enterococcus faecalis* and *E. faecium.* Essential oils (EOs) derived from plants (*Pogostemon heyneanus* and *Cinnamomum tamala*) were also combined with DNases to promote the eradication of established biofilms (Rubini et al., 2018) in methicillin-resistant *Staphylococcus aureus* (MRSA). Proteases were also combined with antiseptics and ethylenediamine tetra-acetic acid (EDTA) for eradication of *S. aureus* and *P. aeruginosa* biofilms (Lefebvre et al., 2016). Other enzymes, such as proteinase K (Kumar Shukla and Rao, 2013), glycoside hydrolases (Baker et al., 2016) and α-amylase (Watters et al., 2016) were also investigated for their ability to disrupt matrix components. Despite the promising *in vitro* efficacy of these enzymes, further studies in animal models are still required for validation purposes.

(iv) Use of halogenated furanones and inhibitors of QS signaling cascades (Lonn-Stensrud et al., 2009) like acyclic diamine (Kaur et al., 2017). Furanones are known to regulate biofilm formation in bacteria. Studies have indicated that synthetic furanones like (Z)-5-(bromomethylene)furan-2(5H)-one inhibited biofilm formation in *Streptococcus anginosus*, *S. intermedius*, *S. mutans* and *Staphylococcus epidermidis* by interfering with the auto-inducer signalling pathway required for biofilm formation (Lonn--Stensrud et al., 2009). Studies have also indicated that acyclic amines inhibit *S. aureus* biofilm formation by interacting with the Staphylococcal accessory regulator, *SarA* (Arya and Princy, 2013; Balamurugan et al., 2017) and cause a significant reduction in the expression of the *SarA* regulated virulence genes like *fnbA*, *hla* and *hld.* A previous report by Chtchigrovsky et al. (2013) has demonstrated the anti-proliferative activity of *trans-N*-heterocyclic carbene–amine–Pt (II) complexes:m these complexes induce intrastrand cross-links in DNA.

(v) Using natural products like ginseng, garlic extract, usnic acid and azithromycin to inhibit development of bacterial and fungal biofilms (Bjarnsholt et al., 2005; Ding et al., 2018; Wu et al., 2015).

(vi) Use of signaling molecules like nitric oxide (NO) to disperse biofilms. NO is an important factor regulating the dispersal of biofilms containing *P. aeruginosa* (Barraud et al., 2006), *Staphylococcus aureus* (Schlag et al., 2007), as well as those composed of single or multispecies (Barraud et al., 2009a, 2009b). NO-mediated dispersal of biofilm is probably mediated by an increase in phosphodiesterase activity, which results in an overall decrease in intracellular c-di-GMP levels required for biofilm formation (Barraud et al., 2009b).

(vii) CRISPRi (clustered regularly interspaced short palindromic repeats (CRISPR) interference) technology to mitigate QS by inhibiting the luxS gene or fimbriae associated gene (*fim*H) required for biofilm formation (Zuberi et al., 2017a, 2017b). QS precedes biofilm formation and therefore inhibiting biofilm formation by targeting genes involved in QS could be a viable strategy for combating AMR in biofilms. Zuberi et al. (2017a) targeted luxS (involved in the synthesis of Autoinducer-2(AI-2), which is required for the initial stage of biofilm formation) and *fimH* gene (Zuberi et al., 2017b) (required for substratum attachment) and demonstrated that in *E. coli* in which these genes had been knocked down biofilm was inhibited.

(viii) Combating MDR and biofilm formation with antibiotic nanoparticles of antibiotics (Baek and An, 2011; Pelgrift and Friedman, 2013). Nanoparticles made of the metal oxides CuO, NiO, ZnO, and Sb_2O_3 were evaluated for their toxicity to *E. coli*, *Bacillus subtilis* and *Staphylococcus aureus.* CuO nanoparticles were found to be the most toxic to these bacteria (Baek and An, 2011). In a subsequent study it was demonstrated that nanoparticles of antimicrobial agents could overcome drug resistance by decreasing uptake, increasing efflux and inhibiting biofilm formation (Pelgrift and Friedman, 2013).

(ix) Suppression of enzymes associated with cell adhesion, glucan synthesis, acid tolerance, acid production and QS required for biofilm formation (Kulshrestha et al., 2016). In *S. mutans* acid production, acid tolerance and EPS formation are characteristics that majorly contribute to virulence of the biofilm. Calcium fluoride nanoparticles (CaF_2-NPs) inhibited all the above processes to inhibit biofilm formation. Gene expression analysis showed that calcium fluoride nanoparticles induced significant down-regulation of various virulence genes (*vicR*, *gtf*C, *ftf*, *spa*P, *com*DE) associated with biofilm formation.

(X) Use of photodynamic therapy (PDT) to eliminate biofilms (Misba and Khan, 2018; Misba et al., 2016, 2017). PDT generates reactive oxygen species (ROS) in the presence of a photosensitizer, visible light and oxygen and to destroy bacterial cells. The method has been used to kill gram-positive and gram-negative bacteria, as well as inhibit biofilm formation. PDT using a photosensitizer (toluidine blue) conjugated with silver nanoparticles was effective in reducing the viability and biofilm formation *in S. mutans* (Misba et al., 2016). The expression of biofilm related genes was also reduced following PDT. In a subsequent study, phenothiazinium dyes were used to demonstrate that gram-positive bacteria (*Enterococcus faecalis*) are more susceptible to PDT compared to the gram –negative *Klebsiella pneumoniae.* This was shown to occur due to higher levels of ROS generation, better photosensitizer binding efficiency and higher DNA degradation (Misba et al., 2017) that PDT induces in gram-positive bacteria. PDT using phenothiazinium dyes was also effective in inhibiting biofilm formation in *S. mutans* (Misba and Khan, 2018).

(XI) Phage therapy for inhibiting biofilm inhibition (Chaudhry et al., 2017). Lytic phages have been more effective than antibiotics in inhibiting bacterial infections in mice (Alemayehu et al., 2012; Smith and Huggins, 1982). A few studies have also demonstrated that a combination of phage and antibiotic therapies could be very effective in killing *P. aeruginosa* (Lu and Collins, 2009; Torres-Barcelo et al., 2014) and inhibiting biofilm formation (Chan et al., 2016; Danis-Wlodarczyk et al., 2016). Chaudhary et al. (2017) treated *P. aeruginosa* PA14 with two phages and five antibiotics either individually or in combinations with each other; the study found that although individual treatments with either phages or drugs were moderately effective in killing the bacteria, phage–drug combinations were more effective as they exhibited synergistic effect.

13.1. Limitations in hacking strategies

Demonstrating the efficacies of the above strategies is only the first step towards combating biofilm formation in infections and in associated AMR-related issues. Most of these strategies have been limited and restricted to *in vitro* experiments and need to be tried in laboratory animals before they could be assessed at the clinical level. It is possible that some strategies like the use of D-amino acids, halogenated furanones, natural products, signalling molecules and nanoparticles could make the transition from the laboratory to the clinic at least to treat skin disorders. However, the other strategies like phage therapy are more demanding and without successful trials in humans, they are unlikely to be implemented. Gupta (2011) suggested the possibility of evaluating the efficacy of phage therapies as adjuncts to the established antibiotic therapies.

14. The gap areas

This is a comprehensive review on ocular microbes with special reference to biofilm formation and AMR. Though several of these aspects have been studied and documented in detail in non-ocular bacterial isolates there is paucity of data on ocular bacteria. A few of the gap areas include:

(i) Several ocular bacteria are yet to be identified up to the species level.
(ii) Characterisation of ocular microbiomes (conjunctiva, corneal scrapings, ocular fluids, lacrimal etc.) under different physiological conditions and for different ethnicities, climates, geographical regions etc.
(iii) There are no comprehensive systematic studies on ocular biofilm formation, including the potential and optimum conditions for biofilm formation, relationship between biofilm formation and AMR etc.
(iv) There are few studies on the molecular basis of AMR and biofilm formation using proteomics, DNA-microarray and candidate gene mutation approaches for ocular microbial isolates.
(v) There are few studies that have carried out in-depth investigations on the relationships between different genera and species of bacteria in a biofilm.
(vi) There is a need for screening synthetic and naturally occurring compounds for dispersing biofilms.

15. Conclusion

Microbiome studies are crucial for furthering our understanding of the roles that different the microbiota play in the onset and progression of several diseases including ocular diseases. Furthermore, AMR is a global phenomenon that needs to be addressed by international health organisations and individual nations and communities so that effective strategies are implemented to curtail the burden of the disease.

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